Matthew L. McCoy

Additive Manufacturing Technical Lead | Machine Architecture & Process Development | Product Launch & Scale

NSF Graduate Research Fellow | Atlanta, GA

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I build and architect industrial powder-bed additive manufacturing systems from first principles and lead multidisciplinary engineering teams that turn them into reliable, scalable products. At Formlabs, the world's largest supplier of professional SLA and SLS systems, I became a printer-wide technical authority in SLS R&D across materials, process physics, optics, thermals, inert-gas handling, hardware, controls, calibration, and software. I held engineering sign-off authority for the 30+ engineer Fuse 1+ 30W launch, owning process architecture, requirements, verification, factory software, and machine acceptance; the Fuse 1/1+ family now represents over 55% of SLS printers sold worldwide. I then returned to lead early Fuse X1 optical and thermal architecture. For my Ph.D., I built the first working platform for Radio Frequency Additive Manufacturing (RFAM), a volumetric polymer powder-bed process that sinters a whole part at once instead of rastering it, together with its multiphysics design software and fully matched RF heating network. **I bring the technical depth to make system-level decisions and the leadership to drive complex printer programs through production.**

SELECTED PROOF

- **Engineering authority:** led technical execution for a 30+ engineer program and 130+ ECOs; owned engineering sign-off for launch gates, factory acceptance, and shipment release on the fastest product launch in Formlabs history.
- **Printer-wide technical depth:** developed SLS materials, process physics, optics, thermal and inert-gas systems, hardware, controls, calibration, and factory and customer software.
- **First-principles range:** built a novel powder-bed process's machine, multiphysics design software, and RF power-delivery network from scratch, entering RF engineering with no prior domain background.

EXPERIENCE

Formlabs — May–Sep. 2019; May 2021 – Sep. 2023

Selective Laser Sintering Research & Development — *Somerville, MA*

Senior Optical and Thermal Systems Engineer (P3) — *Apr. 2023 – Sep. 2023*

- Owned early Fuse X1 optical architecture; reduced the laser, spot-size, hatch, material, smoke, and thermal design space to a few characterized options and presented the system recommendation to engineering leadership and the C-suite.
- Translated process physics into requirements, verification, and executive trade studies; contributed patent disclosures, mentored and hired engineers, and was later offered technical leadership of a five-engineer product-incubation team.

Print Process Engineer (Core R&D) & Technical Program Manager (P2) — *May 2021 – Apr. 2023*

- Engineering authority for the 30+ engineer Fuse 1+ 30W redesign and 130+ ECOs; approved EVT/DVT/PVT gates and production release in nine months. Fuse 1/1+ now represents over 55% of SLS printers sold worldwide.
- Owned process architecture, requirements, verification, factory-test software, and acceptance of contract-manufactured machines in China; trained factory teams, debugged launch issues across time zones, and held shipment-release authority.
- Developed SLS materials, process parameters and controls, upgraded thermal architecture, and a recirculating inert-gas system; created calibration tools that cut dimensional error 3× (±300 to ±100 μm) and enabled LEGO's global Fuse rollout.

Mechanical Engineering & Print Process Intern, Fuse SLS — *May 2019 – Sep. 2019*

- Resolved a laser-coordinate failure that had become a critical beta-testing bottleneck for part fitment; improved pre-launch Fuse geometry accuracy 25% through laser and calibration studies.
- Designed core Fuse Sift powder-processing functions and built the first working Fuse Blast automated media-blasting prototype, selected for later commercialization.

Georgia Institute of Technology – RFAM Platform Architect & Lead Developer — Sep. 2023 – Present

Ph.D. Candidate, Digital Design and Manufacturing Lab / NSF GRFP — *Atlanta, GA*

- Built the first working RFAM platform, a volumetric polymer powder-bed process that patterns conductive dopant and sinters the entire part at once with RF energy; secured internal research funding and a fully funded three-year NSF Graduate Research Fellowship (~$170K in stipend, tuition, and fees), and owned machine architecture, powder handling, high-viscosity jetting, controls, instrumentation, and safety systems.
- Designed and built the 13.56 MHz high-voltage RF power network from scratch: custom-wound ferrite step-up transformers, HT50 ceramic capacitor banks, impedance matching, and a Faraday enclosure. Reaching a stable match into a capacitive, lossy, continuously shifting load was non-trivial; converting that match into efficient energy transfer through the powder bed was harder still, and took 38 physical network builds guided by a fitted physics model that matched hardware within 2–3% and was released as open-source software.
- Melted and densified a graphite-doped nylon cube in under ten minutes, against the hours a conventional laser powder-bed fusion cycle requires; sustained 300 W at 0.17% reflected energy across a 138.5 kJ run while holding the match by hand as the load drifted through melt.
- Root-caused a common-mode RF leakage path corrupting in-situ thermography, cutting coax common-mode current 25 dB and restoring the camera to its RF-off noise floor through 300 W.
- Wrote and finite-difference-verified HEATR (High-frequency Electrothermal Additive Thermal Resolver), a custom solver coupling electrothermal, phase-change, and densification physics, with an adjoint pipeline that converts part geometry into filtered, 4-bit printable material maps; graded designs outperformed uniform material on every benchmark geometry and beat the best prior compensation method in 72% of cases.
- Co-authored peer-reviewed WAAM and hybrid WAAM/powder-blown DED research, including a 2025 *npj Advanced Manufacturing* paper; three journal manuscripts in preparation.

EDUCATION

Georgia Tech Ph.D. Mechanical Engineering | Minor: Industrial Design | NSF Graduate Research Fellow — **Sep. 2023 – Feb. 2027**

Dissertation: Advancing Volumetric Additive Manufacturing via Radio Frequency Energy Delivery: Process Development & Characterization

Northern Illinois University B.S. Mechanical Engineering | Summa Cum Laude — **Aug. 2017 – May 2021**

Minor: Music | GPA: 3.95/4.00 | DeKalb, IL

EXTENDED PROFESSIONAL NARRATIVE & CAREER HIGHLIGHTS

Formlabs: Technical Leadership and Flagship Product Launch

Matthew McCoy joined Formlabs immediately after completing his undergraduate degree as a Print Process Engineer on the Selective Laser Sintering (SLS) research and development team. Within three months he assumed Technical Program Management ownership of the Fuse 1+ 30W platform, coordinating more than 30 engineers across mechanical, electrical, firmware, and software teams while managing over 130 engineering change orders through the full new product introduction cycle. In parallel he managed the factory test software team, which defined and validated more than 150 acceptance tests gating every shipment.

Partnered with Operational Program Management to deliver the company's fastest printer launch, prototype to full production in nine months. The Fuse 1+ 30W became Formlabs' flagship SLS system and the best-selling SLS printer in the world, capturing roughly 55% of the global SLS market by unit volume and deploying thousands of systems across professional and industrial environments.

Within two years McCoy was promoted to Senior Optical and Thermal Systems Engineer and Technical Lead for the company's next-generation SLS platform, now shipping as the Fuse X1. During early concept development, when uncertainty around architecture stalled program direction, he independently initiated and executed an in-depth technical study of the program's highest-risk subsystem, the optical system. That investigation produced the first concrete technical foundation for the next-generation platform and directly catalyzed the internal decision to greenlight development.

Formlabs: Field-Deployed Diagnostic Systems for Precision Manufacturing

A central challenge for professional SLS is environmental variability: factory calibration cannot account for geographic and seasonal thermal shifts that alter powder behavior and part accuracy in the field. McCoy developed two customer-operated diagnostic systems in response. The first let untrained operators reach calibration precision that previously required on-site technician visits costing \$5,000 to \$10,000 per call; built around \$15 consumer-grade calipers and purpose-designed artifacts, it cut dimensional error threefold, from roughly $\pm 300\,\mu$m to $\pm 100\,\mu$m, and directly enabled LEGO's global Fuse deployment. The second used engineered geometric features and torque-driven "spinners" embedded in printed artifacts to reverse-engineer print parameters and detect thermal drift inside the machine. Both remain in active use more than three years on, and were recently featured by Adam Savage in a video on the Fuse platform.

NSF-Funded Doctoral Research at Georgia Tech: Radio Frequency Additive Manufacturing

McCoy is completing his Ph.D. in Mechanical Engineering with a minor in Industrial Design at the Georgia Institute of Technology, supported by the National Science Foundation Graduate Research Fellowship Program. His dissertation develops Radio Frequency Additive Manufacturing (RFAM), a volumetric polymer powder-bed process that replaces raster laser scanning with patterned dopant deposition followed by volumetric radio-frequency heating, densifying parts through electromagnetic coupling rather than surface conduction. It is one of very few volumetric additive processes demonstrated anywhere, and the only one operating on thermoplastic powder rather than photopolymer resin. Working under Dr. Carolyn Seepersad, Chair of Georgia Tech's School of Mechanical Engineering, he won internal research funding on the strength of the concept pitch and built the first working RFAM platform from the ground up within a single year.

The system integrates sub-100 μm motion control, high-viscosity inkjet deposition capable of suspending 50 wt% dopant loading, a 600 W 13.56 MHz RF energy delivery chain with Faraday isolation, and a scanner-based diagnostic platform that quantifies dimensional accuracy and dopant distribution, classifying ink concentration to 98.9% accuracy with Bayesian Gaussian Mixture Models.

Designing the RF Power Delivery System

Delivering usable power into a graphite-doped polymer powder bed proved to be the central engineering obstacle. The load is capacitive, lossy, and shifts continuously as the material heats and melts, so a match that holds at room temperature walks away as the part densifies. McCoy entered RF engineering with no prior background in the field and designed and built the power network from scratch: custom-wound ferrite step-up transformers, high-voltage ceramic capacitor banks, impedance matching, forced-air cooling, and a Faraday enclosure. He iterated 38 physical network builds, fitting a transformer model across them that located the governing secondary-side resonance and explained why earlier topologies had failed. The model predicted node voltage and resonant frequency within 2 to 3% of hardware measurement, and the matching-network design tool built around it was released as open-source software.

The resulting network sustains 300 W at 13.56 MHz with 0.17% of delivered energy reflected across a 138.5 kJ run, holding the match through melt as the load drifts. It melts and densifies a graphite-doped nylon cube in under ten minutes, a cycle a conventional laser powder-bed fusion machine needs hours to complete. Along the way he root-caused an RF common-mode leakage path that was corrupting in-situ thermography, restoring the thermal camera to its RF-off noise floor through 300 W and cutting coax common-mode current by 25 dB.

To direct the process, McCoy wrote HEATR (High-frequency Electrothermal Additive Thermal Resolver), a custom solver coupling electrothermal, phase-change, and densification physics, verified against finite-difference gradients. Its adjoint pipeline inverts part geometry into manufacturable, 4-bit dopant maps; graded designs outperformed uniform material on every benchmark geometry tested.

Research Direction

RFAM targets several structural limits of conventional SLS: it avoids the powder degradation that caps reuse at five to seven cycles, directs energy toward part formation rather than heating an entire bed, decouples throughput from scanning, and projects to machines an order of magnitude below the \$250,000 typical of industrial platforms. The broader direction is computationally controlled radio-frequency energy delivery: RF heating is used industrially as a bulk process, and steering it with spatial fidelity opens a design space in which fields, composition, and geometry are co-designed to place energy inside a structure.